

HS-GPBoost: A Spatial–Feature Framework for Explainable County-Level Health Prediction

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Abstract. Machine learning models such as tree ensembles and gradient boosting are widely used in public-health analytics due to greater interpretability than deep learning. However, these models assume county independence, which fails for regional health outcomes exhibiting strong spatial dependence, leading to biased predictions that do not reflect how risk varies across regions. This work addresses this limitation with HS-GPBoost, a spatial-feature learning framework based on Gaussian Process Boosting that combines nonlinear feature effects with a spatial Gaussian Process, capturing unobserved regional structure while providing an interpretable decomposition of feature-driven and geographic variation. We benchmark the framework against linear, tree-based, spatial autoregressive, and Gaussian process baselines across ten county-level health outcomes from the 2024 County Health Rankings under both random and spatial cross-validation, and include a held-out Florida case study to examine spatial prediction and uncertainty behavior. Results confirm that modeling spatial dependence directly is essential for accurate and interpretable regional health prediction.

Keywords: Spatial machine learning · Gaussian processes · GPBoost · explainable AI · public health · uncertainty quantification.

1 Introduction

Health outcomes in the United States vary widely across regions, as shown in Figure 1. Counties that are geographically close often share similar socioeconomic conditions, environmental exposures, and behavioral risk patterns, which create clear spatial clusters in chronic disease, mental distress, and premature mortality. National datasets such as the County Health Rankings (CHR) provide rich multivariate information, yet producing reliable county-level predictions remains difficult because most machine learning models implicitly assume that counties are independent.

Traditional feature-only models generate predictions that shift abruptly across county borders because they do not capture the spatial processes that drive regional health outcomes, leading to biased estimates and boundary artifacts.

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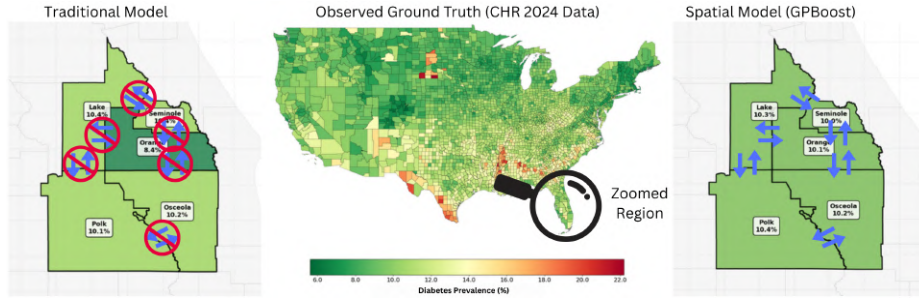


Fig. 1. Traditional models ignore spatial dependence and yield fragmented predictions (left), while HS-GPBoost produces smoother, spatially coherent estimates (right). The center map shows the spatial structure in CHR 2024

Spatially aware approaches can address this, but deep spatial architectures such as graph neural networks introduce substantial training cost and complexity. Gaussian Process Boosting (GPBoost) offers an efficient alternative by combining gradient-boosted trees with a spatial Gaussian process in a single probabilistic framework, capturing both nonlinear feature effects and latent geographic structure without requiring GPU-intensive training.

A further requirement in public-health applications is interpretability. Policymakers need to know which factors drive regional risk and where predictions are less certain. Feature-only models attribute all variation to observed predictors, offering no insight into unmeasured geographic processes. By separating feature-driven effects from spatial residuals, GPBoost provides an interpretable decomposition that conventional models cannot.

This work presents a systematic spatial–feature learning framework that applies GPBoost to county-level health prediction across ten outcomes from the 2024 County Health Rankings. We evaluate the framework against both non-spatial baselines and classical spatial models under rigorous spatial cross-validation, and we analyze its interpretability and uncertainty behavior.

The contributions are as follows. **(1) We design and evaluate a spatial–feature learning framework** based on HS-GPBoost that jointly models nonlinear feature effects and latent spatial structure for county-level health prediction, benchmarked against linear, tree-based, spatial autoregressive, and Gaussian process baselines under both random and spatial cross-validation. **(2) We provide an interpretability analysis** combining gain-based feature importance with spatial uncertainty estimates to identify key drivers of health disparities and flag regions where predictions are less reliable. **(3) We evaluate the framework across ten county-level health outcomes** from the 2024 County Health Rankings using disease-specific feature sets that follow epidemiological guidelines. To our knowledge, no prior work benchmarks GPBoost across multiple County Health Rankings outcomes with spatial cross-validation. **(4) We include a held-out Florida case study** to examine spatial prediction patterns, uncertainty behavior, and out-of-region generalization.

2 Related Work

Spatial Health Patterns. Prior studies consistently show that U.S. county-level health outcomes follow strong geographic patterns and incorporating spatial structure improves predictive performance [2][7]. Some works also [4] highlight how socioeconomic and environmental factors create localized clusters of chronic disease. **Explainable Machine Learning for Health.** Global feature importance has been used to identify key behavioral and socioeconomic drivers of county-level outcomes [2] while spatial heterogeneity has been effective in lung cancer mortality prediction[1]. Furthermore, SHAP analyses have also been applied for post-hoc explainability for health outcome prediction [5,6]. However, these methods provide feature-based explanations only and do not address latent spatial processes that often dominate regional health variation.

Spatial Machine Learning with GPBoost. Recent work introduces GPBoost as a framework that combines gradient boosting with Gaussian processes to capture nonlinear feature effects and spatial correlation [11] and it has been effective for multiple domains [13]. Despite these advances, prior studies do not apply GPBoost to County Health Rankings outcomes or evaluate its behavior across multiple health indicators. No existing work integrates spatial random effects, feature importance, spatial uncertainty, and out-of-distribution regional assessment within a unified spatial-feature modeling framework.

3 Methodology

3.1 Problem Formulation

We define the spatial health prediction task using the dataset $\mathcal{D} = (x_i, s_i, y_i)_{i=1}^N$, where each county c_i has a feature vector $x_i \in \mathbb{R}^p$, centroid coordinates $s_i = (\text{lat}_i, \text{lon}_i) \in \mathbb{R}^2$, and an observed health outcome $y_i \in \mathbb{R}$. The number of counties N ranges from 1,951 to 3,143 depending on data availability. All outcomes are taken from the 2024 County Health Rankings. Prevalence-based outcomes (e.g., diabetes, mental distress) are expressed as percentages, while mortality-based outcomes (e.g., premature death, injury deaths) are age-adjusted rates per 100,000 residents. The goal is to learn a function $f : \mathbb{R}^p \times \mathbb{R}^2 \rightarrow \mathbb{R}$ that maps features and geographic location to a predicted health outcome $\hat{y}_i = f(x_i, s_i)$.

3.2 Model Architecture Overview

Figure 2 shows the Hybrid Spatial GPBoost (HS-GPBoost) architecture, which integrates gradient-boosted trees with a spatial Gaussian process in a unified framework. The model takes feature vectors $\mathbf{x}_i$ and geographic coordinates $\mathbf{s}_i$ as inputs. The boosting component $F(\mathbf{x}_i)$ captures nonlinear feature effects, while the Gaussian process $g(\mathbf{s}_i)$ models spatial dependence using the exponential kernel described in Section 3.5. Together, they represent both observed predictors and latent spatial structure. The model outputs point predictions $\hat{y}_i = F(\mathbf{x}_i) + g(\mathbf{s}_i)$, predictive uncertainty $\sigma_{\text{pred}, i}$, and feature importance scores from the tree ensemble.

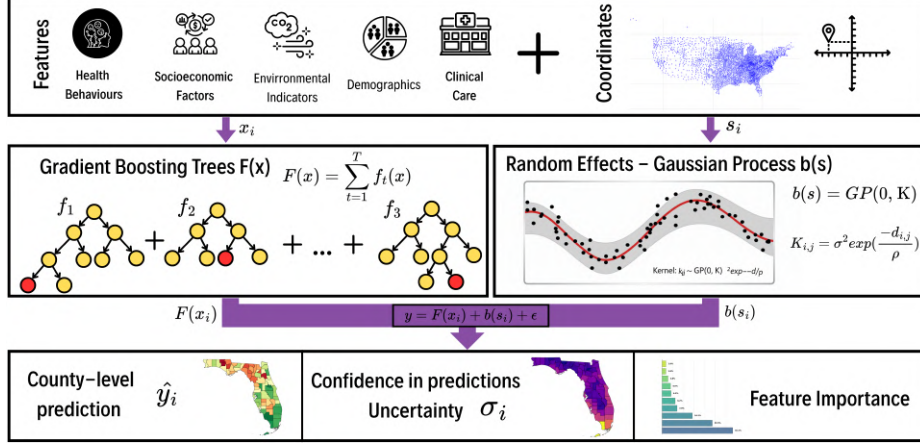


Fig. 2. HS-GPBoost architecture showing gradient boosting with a spatial Gaussian process to generate predictions, uncertainty, and feature importance.

3.3 Model Decomposition

County-level health outcomes depend on both observed determinants and latent geographic influences. Using an additive structure, we write

$$y_i = F(\mathbf{x}_i) + g(\mathbf{s}_i) + \epsilon_i, \quad (1)$$

where $F(\mathbf{x}_i)$ represents feature-driven effects, $g(\mathbf{s}_i)$ is a Gaussian-process spatial term with covariance $k(\mathbf{s}_i, \mathbf{s}_j)$, and ϵ_i is independent noise.

3.4 Spatial Autocorrelation

County-level health outcomes show clear spatial autocorrelation, where nearby counties tend to share similar health burdens due to unobserved regional factors not captured by available features. We model this dependency using a stationary exponential covariance kernel:

$$k(\mathbf{s}_i, \mathbf{s}_j) = \sigma^2 \exp\left(-\frac{d_{ij}}{\rho}\right), \quad (2)$$

where σ^2 controls spatial variance, d_{ij} is the geodesic distance between county centroids and ρ determines the correlation range. This formulation captures latent geographic influences that remain after accounting for socioeconomic and behavioral predictors.

3.5 Gaussian Process Regression

To benchmark spatial dependence independent of feature-based learning, we implement Gaussian Process Regression (GPR), modeling the outcome as

$$f(\mathbf{s}) \sim \mathcal{GP}(\mu(\mathbf{s}), k(\mathbf{s}, \mathbf{s}')), \quad (3)$$

$$y_i = f(\mathbf{s}_i) + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, \sigma_n^2). \quad (4)$$

The covariance function $k(\mathbf{s}_i, \mathbf{s}_j)$ uses the exponential kernel from Eq. 2, which encodes the tendency for nearby counties to have similar health outcomes. Hyperparameters (σ^2, ρ) are estimated by maximizing the log marginal likelihood. GPR, a fully probabilistic spatial model, is ideal for direct uncertainty quantification. However, due to high computational cost $\mathcal{O}(n^3)$ and the absence of nonlinear feature effects, it serves as a purely spatial baseline for comparison with the hybrid GPBoost model [9].

3.6 Gradient Boosting for Feature Learning

To isolate the contribution of feature-based learning, a non-spatial variant of GPBoost [11], removing the Gaussian process term, models the outcome as

$$y_i = F(\mathbf{x}_i) + \epsilon_i, \quad (5)$$

where $F(\mathbf{x}_i) = \sum_{t=1}^T f_t(\mathbf{x}_i)$ is an additive ensemble of regression trees trained by gradient boosting, and $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$ is independent noise. This corresponds to GPBoost with $g(\mathbf{s}_i) = 0$, retaining the same likelihood framework but without spatial effects. We refer to this feature-only variant as **HS-GPBoost-** to isolate the contribution of gradient boosting from spatial modeling. It serves as an ablation baseline, allowing us to quantify the specific performance gain attributable to the Gaussian process component. At each iteration, a weak learner f_t is fit to the residuals

$$r_i^{(t)} = y_i - F_{t-1}(\mathbf{x}_i), \quad (6)$$

using gradient-based optimization of the loss (e.g., squared error). This approach captures nonlinear relationships among socioeconomic and behavioral predictors but assumes counties are independent, ignoring any spatial autocorrelation.

3.7 Joint Spatial–Feature Learning Framework

Model Formulation The gradient boosting and Gaussian process components are combined within a single probabilistic optimization framework [11]. Each county-level health outcome y_i is expressed as:

$$y_i = F(\mathbf{x}_i) + g(\mathbf{s}_i) + \epsilon_i, \quad (7)$$

Here, $F(\mathbf{x}_i)$ captures nonlinear feature relationships via an additive ensemble of regression trees, while $g(\mathbf{s}_i)$ represents latent spatial processes structured by the covariance kernel.

Joint Likelihood and Objective Function: Given n observations $\mathbf{y} = (y_1, \dots, y_n)^\top$, the joint model assumes:

$$\mathbf{y} = F(\mathbf{X}) + \mathbf{g} + \boldsymbol{\epsilon}, \quad \mathbf{g} \sim \mathcal{N}(\mathbf{0}, \mathbf{K}), \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma_\epsilon^2 \mathbf{I}),$$

where $\mathbf{K}$ is the $n \times n$ covariance matrix with entries $K_{ij} = k(\mathbf{s}_i, \mathbf{s}_j)$. The marginal likelihood of $\mathbf{y}$ conditioned on $F(\mathbf{X})$ and kernel parameters $\theta = (\sigma^2, \rho, \sigma_\epsilon^2)$ is:

$$p(\mathbf{y} \mid F, \theta) = \mathcal{N}(\mathbf{y} \mid F(\mathbf{X}), \mathbf{K}_\theta + \sigma_\epsilon^2 \mathbf{I}). \quad (8)$$

Maximizing this marginal likelihood is equivalent to minimizing the following negative log-likelihood objective:

$$\mathcal{L}(F, \theta) = \frac{1}{2}(\mathbf{y} - F(\mathbf{X}))^\top (\mathbf{K}_\theta + \sigma_\epsilon^2 \mathbf{I})^{-1} (\mathbf{y} - F(\mathbf{X})) + \frac{1}{2} \log |\mathbf{K}_\theta + \sigma_\epsilon^2 \mathbf{I}|. \quad (9)$$

This objective jointly optimizes tree-based learner $F(\mathbf{X})$ and the Gaussian process parameters θ , balancing the feature-driven fit with spatial smoothness. The training procedure alternates between two optimization steps: (1) fitting gradient-boosted trees on residuals

$$r_i^{(m)} = y_i - g^{(m-1)}(\mathbf{s}_i).$$

, and (2) spatial covariance parameters $\theta = (\sigma^2, \rho, \sigma_\epsilon^2)$ are re-estimated by maximizing the restricted log-marginal likelihood:

$$\theta^{(m)} = \arg \max_{\theta} \left[-\frac{1}{2} (\mathbf{r}^{(m)})^\top (\mathbf{K}_\theta + \sigma_\epsilon^2 \mathbf{I})^{-1} \mathbf{r}^{(m)} - \frac{1}{2} \log |\mathbf{K}_\theta + \sigma_\epsilon^2 \mathbf{I}| \right]. \quad (10)$$

This alternating process ensures that $F(\mathbf{x}_i)$ captures feature-level structure, while $g(\mathbf{s}_i)$ isolates spatial dependencies orthogonal to measured covariates.

Spatial Prediction and Uncertainty Quantification: After training, the posterior mean and covariance for a new county with features $\mathbf{x}_*$ and location $\mathbf{s}_*$ are given by:

$$\hat{y}_* = F(\mathbf{x}_*) + \mathbf{k}_*^\top (\mathbf{K}_\theta + \sigma_\epsilon^2 \mathbf{I})^{-1} (\mathbf{y} - F(\mathbf{X})), \quad (11)$$

$$\sigma_{\text{pred},*}^2 = k(\mathbf{s}_*, \mathbf{s}_*) - \mathbf{k}_*^\top (\mathbf{K}_\theta + \sigma_\epsilon^2 \mathbf{I})^{-1} \mathbf{k}_*, \quad (12)$$

where $\mathbf{k}_*$ is the covariance vector between the new location $\mathbf{s}_*$ and the training set. Point predictions $\hat{y}_*$ and predictive intervals $\hat{y}_* \pm 1.96 \sigma_{\text{pred},*}$ provide both mean estimates and spatially explicit uncertainty.

4 Experiments

4.1 Data

We use the 2024 County Health Rankings & Roadmaps dataset (CHR) [12,10], covering health behaviors, clinical care, socioeconomic factors, and physical environment indicators for all U.S. counties, indexed by FIPS codes. We model 10 outcomes and each one of them is paired with an epidemiologically motivated predictor set of 8–11 features (e.g., Table 1) with careful exclusion of rows with missing values, invalid FIPS code or missing geographic coordinates. The sample size ranges from 1,951 to 3,143 counties based on the specific outcome. All predictor variables, outcome variables, and spatial coordinates were standardized using z-score normalization. County centroids are obtained via a FIPS-based join from the *USA Counties Coordinates* dataset [8].

Table 1. Example predictor set for Diabetes Prevalence.

Feature	Description
Adult Obesity	Adults with BMI ≥ 30 (%)
Physical Inactivity	No leisure-time physical activity (%)
Food Environment Index	Food access & affordability (0–10)
Exercise Opportunities	Access to recreation facilities (%)
Median Income	Median household income (\$)
High School Completion	Adults 25+ with diploma (%)
Adult Smoking	Current smokers (%)
Excessive Drinking	Binge or heavy drinking (%)

4.2 Baseline Models

To compare our framework with prior public-health modeling approaches, we implement several representative baselines. **Allen (2023) [2]** uses linear regression to model county-level obesity, which we reproduce using socioeconomic predictors. **Feng and Jiao (2021) [4]** apply decision trees to estimate neighborhood-level prevalence of chronic diseases; we adapt this approach to the county-level outcomes in our study. **Athanasίου et al. (2020) [3]** develop an explainable XGBoost model for cardiovascular risk prediction, which we use as a strong nonlinear baseline. For our primary comparison, we evaluate the **Hybrid Spatial GPBoost (HS-GPBoost)** model, which integrates gradient-boosted trees with a spatial Gaussian Process to jointly capture nonlinear feature effects and latent geographic structure, extending the GPBoost framework of [11] to county-level health prediction. We also include a **Spatial Autoregressive model (SAR)** as a classical spatial econometrics baseline that captures spatial dependence through a spatially lagged dependent variable, allowing us to compare GPBoost’s Gaussian process formulation against established areal spatial models.

4.3 Spatial Cross-Validation

To evaluate geographic generalization, we use a 5-fold spatial cross-validation scheme. Counties are first connected via a Queen-contiguity adjacency graph and then partitioned into five spatially compact clusters using METIS. In each fold, one cluster serves as the test region, and training counties exclude all test counties and their immediate neighbors, preventing spatial leakage. Metrics are averaged over the five folds.

4.4 Results

Random CV Performance. Under 5-fold random CV (Tables 2–3), most models show high accuracy for physical health outcomes. HS-GPBoost performs best across all ten outcomes, reaching R^2 values above 0.90 for Poor or Fair Health, Frequent Physical Distress, Poor Physical Health Days, and Diabetes. XGBoost is competitive for physical outcomes but declines sharply for mental health and mortality. Linear regression and decision trees perform poorly, reflecting their limited ability to capture nonlinear or interaction effects.

Table 2. Random CV: Test Metrics for Linear and Tree-Based Models (LR, DT, XGB).

Disease	LR Allen (2023)			DT Feng (2021)			XGB Athanasiou (2020)		
	R^2	RMSE	MAPE	R^2	RMSE	MAPE	R^2	RMSE	MAPE
Poor Fair Health	0.915	0.013	28.9	0.876	0.016	6.8	0.941	0.011	4.7
Freq Physical Distress	0.854	0.008	19.8	0.832	0.009	5.5	0.911	0.006	4.1
Poor Physical Health	0.774	0.302	18.0	0.746	0.320	6.3	0.860	0.238	4.8
Diabetes Prevalence	0.838	0.009	22.3	0.752	0.011	7.6	0.871	0.008	5.6
Freq Mental Distress	0.581	0.013	12.3	0.481	0.015	6.8	0.705	0.011	5.2
Poor Mental Health	0.415	0.457	11.1	0.272	0.510	7.8	0.578	0.388	5.9
Premature Mortality	0.700	79.342	34.5	0.580	93.958	14.7	0.733	74.857	11.3
Low Birthweight	0.514	0.014	23.8	0.305	0.017	15.9	0.580	0.013	12.5
Injury Deaths	0.453	19.991	28.7	0.123	25.296	20.4	0.456	19.931	16.2
Drug Overdose Deaths	0.159	14.994	57.5	-0.335	18.822	54.1	0.157	15.001	45.7

Table 3. Random CV: Test Metrics for Gaussian Process Models (HS-GPBoost–, GP, HS-GPBoost).

Disease	HS-GPBoost–			GP			HS-GPBoost		
	R^2	RMSE	MAPE	R^2	RMSE	MAPE	R^2	RMSE	MAPE
Poor Fair Health	0.909	0.013	6.1	0.663	0.026	11.1	0.962	0.009	3.6
Freq Physical Distress	0.874	0.007	5.0	0.658	0.012	7.9	0.957	0.004	2.6
Poor Physical Health	0.817	0.272	5.7	0.695	0.351	7.0	0.942	0.153	3.0
Diabetes Prevalence	0.822	0.009	6.8	0.613	0.014	9.2	0.914	0.006	4.4
Freq Mental Distress	0.662	0.012	5.7	0.716	0.011	5.0	0.861	0.008	3.4
Poor Mental Health	0.525	0.412	6.4	0.694	0.331	4.9	0.790	0.274	4.1
Premature Mortality	0.704	78.747	12.2	0.502	102.290	16.3	0.725	76.007	11.3
Low Birthweight	0.551	0.014	13.0	0.569	0.014	12.5	0.666	0.012	11.0
Injury Deaths	0.425	20.534	16.7	0.357	21.692	17.9	0.579	17.561	14.0
Drug Overdose Deaths	0.221	14.431	45.9	0.501	11.546	33.8	0.571	10.708	31.1

Spatial CV Performance. Spatial 5-fold CV (Table 4) produces lower R^2 values due to enforced geographic separation, yet HS-GPBoost remains the top model in eight of ten outcomes. On average, spatial CV reduces HS-GPBoost performance by 0.11 in R^2 , compared with drops of 0.07 for XGBoost, 0.04 for linear regression, and 0.08 for decision trees. The SAR baseline captures some spatial structure but cannot model nonlinear feature effects, and it generally performs between LR and XGBoost across most outcomes.

Physical health outcomes show the strongest spatial CV accuracy: Poor or Fair Health ($R^2 = 0.921$), Frequent Physical Distress ($R^2 = 0.904$), Poor Physical Health Days ($R^2 = 0.890$), and Diabetes Prevalence ($R^2 = 0.836$). These align well with well-measured behavioral and socioeconomic predictors.

Mental health outcomes achieve moderate accuracy, with HS-GPBoost reaching $R^2 = 0.781$ (Frequent Mental Distress) and $R^2 = 0.704$ (Poor Mental Health Days). Spatial modeling yields large gains over GPB–, indicating strong unmeasured geographic influences.

Mortality outcomes are more difficult. Premature Mortality achieves moderate accuracy ($R^2 = 0.671$), while Injury Deaths ($R^2 = 0.411$) and Drug Overdose Deaths ($R^2 = 0.404$) remain challenging across all models, reflecting fast-changing or sparsely measured underlying processes.

4.5 Spatial Effects

Comparing HS-GPBoost with HS-GPBoost– isolates the contribution of spatial structure. Spatial gains are small for physical health (e.g., $\Delta R^2 = +0.047$ for

Table 4. Spatial CV: Test Metrics for LR, DT, XGB, SAR, and HS-GPBoost Models.

Disease	Model	R ²	RMSE	MAPE	Disease	Model	R ²	RMSE	MAPE
Poor or Fair Health	LR	0.911	0.013	29.0	Frequent Physical Distress	LR	0.845	0.008	19.6
	DT	0.844	0.018	7.7		DT	0.784	0.010	6.3
	XGB	0.919	0.013	5.5		XGB	0.869	0.008	4.9
	SAR	0.910	0.013	6.1		SAR	0.843	0.008	5.5
	GPB	0.921	0.013	5.0		GPB	0.904	0.007	3.9
Poor Physical Health Days	LR	0.761	0.310	17.8	Diabetes Prevalence	LR	0.835	0.009	22.6
	DT	0.687	0.355	7.3		DT	0.697	0.012	8.6
	XGB	0.807	0.279	5.7		XGB	0.839	0.009	6.4
	SAR	0.759	0.312	6.5		SAR	0.835	0.009	6.4
	GPB	0.890	0.211	4.1		GPB	0.836	0.009	5.8
Frequent Mental Distress	LR	0.562	0.014	12.3	Poor Mental Health Days	LR	0.397	0.464	11.1
	DT	0.406	0.016	7.2		DT	0.146	0.552	8.3
	XGB	0.648	0.012	5.6		XGB	0.493	0.425	6.5
	SAR	0.544	0.014	6.4		SAR	0.371	0.474	7.3
	GPB	0.781	0.010	4.3		GPB	0.704	0.325	4.8
Premature Mortality	LR	0.688	80.873	34.5	Low Birthweight	LR	0.507	0.014	23.5
	DT	0.505	102.039	16.0		DT	0.241	0.018	17.0
	XGB	0.691	80.515	12.4		XGB	0.528	0.014	13.4
	SAR	0.688	80.918	12.4		SAR	0.460	0.015	13.5
	GPB	0.671	83.138	12.8		GPB	0.594	0.013	12.3
Injury Deaths	LR	0.435	20.299	29.2	Drug Overdose Deaths	LR	0.118	15.358	54.1
	DT	-0.063	27.712	22.4		DT	-0.269	18.399	53.1
	XGB	0.353	21.743	17.7		XGB	0.086	15.639	46.4
	SAR	0.419	20.608	15.9		SAR	-0.058	16.810	38.8
	GPB	0.411	20.744	17.1		GPB	0.404	12.609	35.7

Poor or Fair Health; +0.058 for Diabetes), indicating that measured features explain most variation. Mental health outcomes show much larger gains (+0.159 for Frequent Mental Distress; +0.220 for Poor Mental Health Days), consistent with strong neighborhood-level effects not captured by covariates. For Drug Overdose Deaths, HS-GPBoost substantially outperforms tree-based models, reflecting the spatial diffusion patterns that characterize overdose mortality. SAR shows modest gains over LR for several outcomes but consistently underperforms HS-GPBoost, confirming that the combination of nonlinear feature learning and continuous spatial modeling outperforms linear spatial regression.

4.6 Spatial Prediction: Florida Case Study

Figure 3 shows HS-GPBoost predictions for *Frequent Mental Distress* across Florida’s 67 counties. The model captures the observed north–south gradient, with higher burden in the rural Panhandle and lower rates in coastal South Florida, reflecting both broad regional structure and local county-level variation.

Prediction errors show mild overprediction in several urban counties, including Miami-Dade, and underprediction in parts of the northern interior, suggesting that some urban protective factors and rural stressors are not fully captured by the CHR predictors.

Uncertainty remains low statewide (mean SD 0.7%) but increases along the southern and southeastern coastline, where coastal boundary effects reduce spatial support for the GP component. Larger urban centers also show slightly higher uncertainty due to demographic heterogeneity, resulting in narrow intervals of roughly ± 1.9 percentage points.

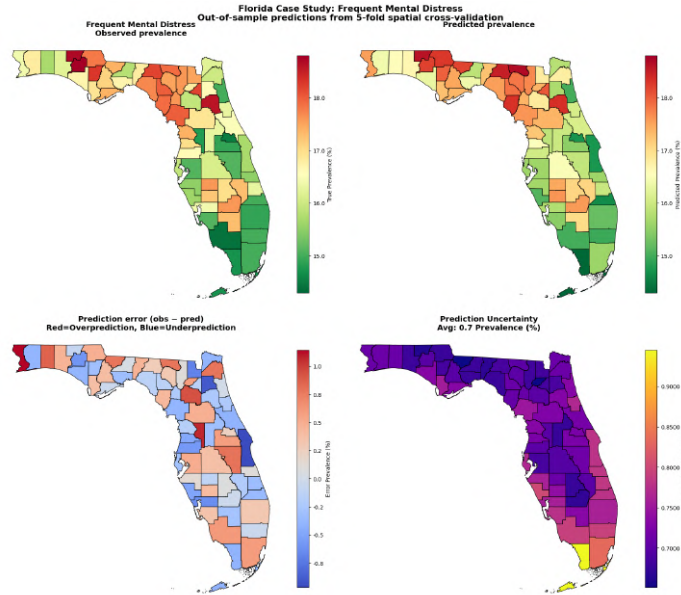


Fig. 3. HS-GPBoost spatial predictions for *Frequent Mental Distress* in Florida. Top: observed vs. predicted prevalence. Bottom: prediction error and predictive uncertainty.

4.7 Feature Importance Analysis

Figure 4 shows gain-based feature importance from HS-GPBoost for six representative outcomes. For physical health outcomes, behavioral risk factors dominate: *Adult Smoking* and *Physical Inactivity* account for the largest share of predictive gain, consistent with established links to chronic disease. Socioeconomic variables such as income and education contribute less, suggesting their effects operate mainly through behavioral pathways.

Mental health outcomes show a different pattern, with socioeconomic conditions as the strongest predictors. *Median Household Income* and *Children in Poverty* carry the most gain, reflecting the influence of economic stress on mental well-being. Healthcare access variables add moderate value, and environmental exposures contribute little across all outcomes.

To assess the stability of gain-based importance, we computed feature rankings independently across all five spatial CV folds. The top-ranked feature remained in the first position across all folds for nine of ten outcomes. For physical health outcomes, *Adult Smoking* and *Physical Inactivity* held ranks one and two in every fold with narrow variation. For mental health outcomes, *Median Household Income* ranked first across all folds. Mortality and birthweight outcomes showed stable leading features as well: *Adult Smoking* for Premature Mortality (rank 1–2 in all folds) and *Children in Single-Parent Households* for Low Birthweight (rank 1–2). Only Drug Overdose Deaths exhibited notable rank variation among the top three features, consistent with the high prediction difficulty for

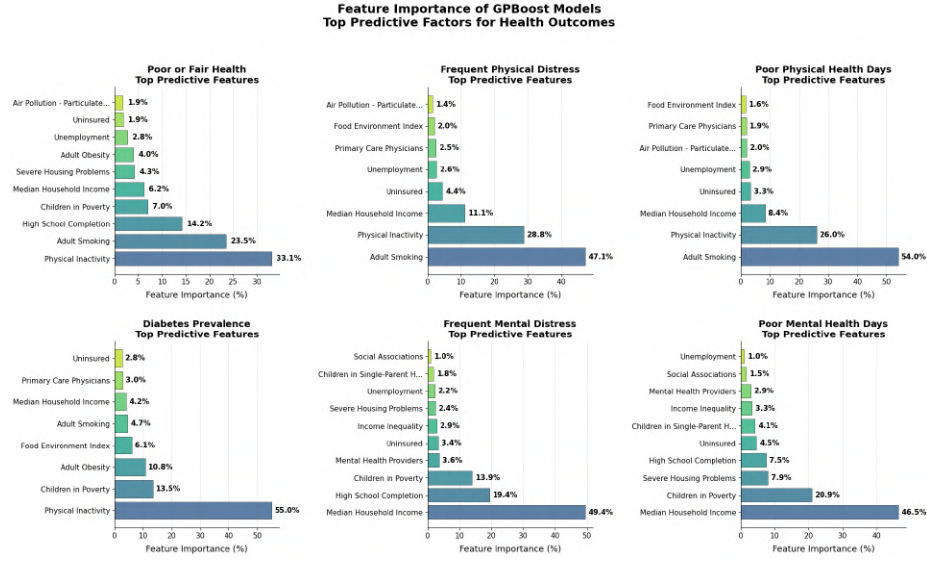


Fig. 4. Gain-based feature importance from HS-GPBoost for six representative health outcomes. Physical health outcomes (top row) are dominated by behavioral risk factors, while mental health outcomes (bottom row) are driven primarily by socioeconomic determinants. Feature contributions are normalized to 100% per outcome.

this outcome ($R^2 = 0.404$). These results indicate that the gain-based rankings reflect stable predictive structure rather than artifacts of particular data splits.

Overall, two patterns emerge: behavioral risks drive physical health variation, and socioeconomic determinants drive mental health outcomes. The concentration of importance in a small set of predictors highlights clear intervention pathways and shows that HS-GPBoost captures domain-aligned signal even in the presence of spatial dependence.

5 Conclusion

In this work, we showed that county-level health outcomes follow strong spatial patterns that traditional feature-only models fail to capture. By combining nonlinear feature effects with a spatial Gaussian Process, HS-GPBoost produced more accurate and geographically coherent predictions across ten outcomes from the 2024 County Health Rankings, outperforming linear, tree-based, and spatial autoregressive baselines under both random and spatial cross-validation. The framework also provided feature importance and spatial uncertainty estimates, offering an interpretable decomposition of behavioral, socioeconomic, and geographic drivers of regional risk. These results demonstrate that modeling spatial dependence directly is essential for accurate and interpretable regional health prediction.

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